

Fauzan Mohammed

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SUMMARY

Experienced **NLP** Engineer with a proven track record in designing and deploying multi-agent LLM systems at scale. Skilled in leveraging **NLP** and **Python** to automate workflows and extract **compliance logic**, ensuring responsible AI compliance. Adept at building intelligent solutions for enterprise platforms, driving measurable improvements in operational efficiency.

EDUCATION

Rensselaer Polytechnic Institute

Masters in IT/CS

Troy, NY

Aug 2023 – Dec 2024

Kwame Nkrumah University of Science and Technology

Bachelor of Science in Mechanical Engineering

Kumasi, Ghana

Sep 2016 – June 2020

EXPERIENCE

Data Scientist – NLP/LLM Systems + Responsible AI

Walmart Global Tech

Aug 2024 – Present

Sunnyvale, CA

- Contributed to the **design** and deployment of a production multi-agent LLM system powering Walmart's conversational shopping platform, orchestrating intelligent search, guided discovery, and personalized recommendations for 20M+ monthly active users.
- Designed** prompt-tuned LLM-as-judge evaluation frameworks for search quality and product relevance assessment across raw and reranked results, driving measurable improvements in ranking precision across product categories.
- Built production guardrail infrastructure using prompt-tuned LLM judges on GCP and Element Gateway, **implementing** real-time safety enforcement and hallucination detection to ensure responsible AI compliance for 10M+ users.
- Designed** and maintained prompt templates, multi-step chains, and persona configurations across agent workflows, **implementing** version-controlled prompt libraries for consistency and reuse across the conversational platform.
- Developed an AI agent-based Event Title Generator leveraging LLM function calling to trigger internal APIs, autonomously infer user intent, map to structured event categories, and generate personalized titles within the shopping experience.
- Architected multi-turn conversation management logic including memory handling, contextual continuity, and dynamic persona assignment across domain-specific agent roles.
- Developed unsupervised **NLP** pipeline using DBSCAN clustering for topic modeling and intent extraction, expanding chatbot query coverage from 15% to 74%.

Data Scientist, Intern

Walmart Global Tech

May 2024 – Aug 2024

Sunnyvale, CA

- Engineered Transformer-based fallback detection system using semantic clustering to classify 100K+ daily fallback queries into 27 intent clusters, improving diagnostic accuracy by 6%.
- Optimized LLM router architecture through prompt engineering and query decomposition, improving task-selection accuracy by 4% and reducing latency by 18%.
- Built automated evaluation dashboards using SQL and **Python** for real-time monitoring of precision, recall, and prompt drift, reducing experimentation cycles by 30%.

Junior Machine Learning Engineer

Omdena Thraive

Dec 2022 – Apr 2023

Remote

- Engineered BERT-based **NLP** models with attention mechanisms achieving 85% classification accuracy on

5K+ mental health journal entries; deployed production analytics dashboard on GCP using Flask and Plotly for 200+ clinical users.

PROJECTS

AI-Powered Financial Rule Automation System

Dec 2024

- Built AI rule-generation engine integrating multiple Transformer models (MiniLM, RoBERTa, DeBERTa) with QuickBooks API, **implementing** multi-task fusion architecture to auto-categorize transactions at 92% precision and reduce manual rule creation by 70%.

LEADERSHIP

Co-founder and Director

Aug 2019 – Present

Fama Foundation

Accra, Ghana

- Led impactful programs across 10 Ghanaian communities, enrolling 325 teenage girls in computer programming and literacy education.

TECHNICAL SKILLS

Languages: Python, SQL, Java

ML/DL: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Sentence Transformers, spaCy, NLTK

LLM & AI: LangChain, LlamaIndex, FAISS, OpenAI/Anthropic/Gemini APIs

Cloud & Data: GCP (Vertex AI, BigQuery, Cloud Run), Docker, Kubernetes, Pandas, NumPy, Spark

MLOps: MLflow, Weights & Biases, Flask, FastAPI, CI/CD, Git, DVC

Specialties: Multi-Agent Systems, LLM Fine-Tuning (SFT, RLHF, DPO), Prompt Engineering, RAG, Model Evaluation, Responsible AI, Function Calling & Tool Use, Conversation Flow **Design**